

Detailed Project Proposal

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7COM1039-0901-2025(Advanced Computer Science Masters Project)

Description

In the field of Data Science, mental health is a topic that has been widely researched. Yet, behavioral and demographical factors that considerably influence mental well-being across diverse populations are still misunderstood and are often confused. The project is inspired by the new findings established by machine learning to predict mental health conditions and aims to enhance such analysis using rather unconventional and less frequently used algorithms. I propose to investigate the key features and their individual importance on a person's mental health levels using machine learning and clustering algorithms applied to a mental health dataset. The analysis will emphasize feature importance and grouping individuals based on their behavioral patterns. The project will also include a real-time interaction by building an interactive visual interface that allows users to understand the correlation among features and analyze their mental health conditions. The project will also address the gaps in previous studies by aggregating related features into composite indicators and will use underutilized machine learning algorithms to understand complex relationships that might be missed by conventional ones. The evaluation will be carried out by using classification metrics such as accuracy, F1-score, AUC, etc., and also supported by visual interpretability tools such as SHAP and Lime, and the outcomes will be compared with previous results.

Project's Title

Predicting Mental Health Risk Levels Using Behavioral Feature Aggregation and Alternative Machine Learning Algorithms with Interactive Visualizations

Aim of the Project

The aim of my project is to predict mental health risk levels by analysing behavioral and demographic factors, merging correlated features into a superior indicator to reveal underlying dynamics, and comparing alternative machine learning algorithms such as CatBoost and LightGBM to surpass the predictions and understandings of the previous studies. Another aim of my project is to build an interactive visualization interface using Streamlit that will enable users to interpret the model's prediction using explainable AI

methods, such as SHAP and Lime-based visual explanation, focusing on the feature importance. The project will also aim to compare the chosen model's performance with studies from previous research using metrics such as accuracy, F1-score, and AUC, to assess whether alternative models have the potential to outperform the conventional ones.

Research Questions/Hypothesis

- Do behavioral and Demographic features contribute to the mental health risk levels of a person?
- Can the integration of behavioral indicators, such as merging 'coping struggles' and 'mood swings' into 'mental strain', improve the ability of machine learning algorithms to predict better?
- Do the alternative algorithms, such as CatBoost and LightGBM, have the potential to outperform the conventional models like Logistic Regression, Random Forest, etc?
- Can Explainable AI methods like SHAP and LIME interpret the most influential indicators contributing to mental health levels, and are the findings aligned with the previous studies?

Objectives

To conduct a literature review on the prediction of mental health using machine learning algorithms, focusing on behavioral and demographic data, and identify gaps and justify the argument for selecting both unconventional and conventional machine learning algorithms for comparison. To Preprocess the selected dataset to handle the missing values, encoding the categorical values to prepare the data to be more suitable for the algorithms, and aggregating features to generate more meaningful indicators. To split the dataset into training and test sets and build several machine learning models, including conventional ones like logistic regression, SVM, and alternative algorithms like CatBoost, LightGBM, for predicting mental health risks. To evaluate and compare the standard models with alternative ones using classification metrics such as accuracy, F1-score, and AUC, etc. To interpret the most influential features contributing to mental health risk levels using Explainable AI methods such as SHAP and LIME. and compare outcomes with past studies to ensure consistency or point out differences. To develop an interactive visualisation interface using Streamlit, allowing users to explore the predictions and also help them interpret the feature importance. To write a complete Project Report documenting the entire research process before the described timeline.

Research Methodology

Research Methodology The project will follow the Waterfall Framework or methodology to build sequentially an interpretable machine learning model that will investigate what key factors affect the quality of the mental health risk level. It will attempt to predict the state of mental health as well. As the Waterfall methodology flows, the project will adapt sequential stages-data collection, preprocessing, model development, model evaluation, and visualization, ensuring the procedure is structured, well-maintained, and the outcome is deliverable. The Data of the project will come from the Mental Health Dataset of Bhavik Jikadara from Kaggle, containing around 292365 anonymous responses. The dataset covers multiple attributes that might have an impact on mental health, including behavioral,

demographic, and psychological attributes such as mood swings, family history, and social stress etc. The Data Preprocessing will include cleaning, encoding categorical variables, and integrating correlated features into a single indicator, such as mood swing and coping struggles might merge into mental strain for better prediction. The project is initially planned to have a test and train split of 70 and 30 ratio; however, this is subject to change depending on the circumstances. Models like Catboost, LightGBM, and Random Forest will be trained, and for better optimisation cross cross-validation will be used. Although the project might adapt TabNet, NGBoost algorithms, if sufficient time is managed. Performance measurement will be done through metrics such as accuracy, F1-score, and AUC as primary choices, with interpretability and computational efficiency being the secondary consideration. Explainable AI Methods, such as SHAP and LIME, will be used to interpret the feature importance of each indicator and visualization of their contribution to the predictions. Most of the Analysis will be conducted by implementing Python Libraries such as scikit-learn, pandas, and streamlit. Validity and reliability will be ensured through the use of stratified k-fold cross-validation, controlled preprocessing, and random state seeding, ensuring the outcomes are consistent. Version Control Git will be implemented to support reproducibility. Ethical standards under the guidance of UK GDPR will be maintained, with anonymization and safe storage. At the last part of the Research, an interactive streamlit interface will be built that allows the user to visualise model predictions, interpret feature contributions, and interact with findings of the model, making the outcomes both transparent and accessible.

Citations

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- Mutalib, S., Shafiee & Abdul-Rahman, S., 2021. Mental Health Prediction Models Using Machine Learning in Higher Education Institutions.
- Xin, Y., Wang, Y., Zhang, X., Li, P., Yang, W., Wang, B., & Yang, J., 2025. Machine learning-based analysis and prediction of factors influencing mental health among children and adolescents in Jiangsu Province.
- Retnaswamy, C., Antony Muthu, A.B., Duraipandi, B. & Manickam, R., 2025. Mental Health Treatment Prediction Using Machine Learning
- Ndikumana, F., Izabayo, J., Kalisa, J., Nemerimana, M., Nyabyenda, E.C., Muzungu, S.H., Komezusenge, I., Uwase, M., Ndagijimana, S., Twizere, C., & Sezibera, V., 2025. Machine learning-based predictive modelling of mental health in Rwandan Youth.